Who Made the Story?

The debate over AI-generated creative work often starts with the wrong question: Was AI used at any point in the creative process? Consider three writers:

- A writer who encounters an AI-generated definition while searching online for a word.
- A writer who uses an AI assistant for occasional feedback.
- A writer who asks AI to generate an entire novel and publishes the result unchanged.

A simple yes-or-no test for AI involvement risks treating these three situations alike, even though the contributions AI makes to the finished work could hardly be more different.

A more useful question is how much of the work each participant actually contributed, and what resources were required to produce it.

I propose evaluating creative authorship along two axes: the amount and significance of the content contributed by each participant, and the energy cost associated with producing those contributions. These axes would not capture every possible dimension of creative work, but they offer a practical starting point for distinguishing human-made, hybrid, and AI-made works.

Two axes of creative authorship

The first axis measures creative contribution. In the case of writing, this cannot simply mean counting words. The number of words produced by a participant tells us something about their contribution, but not necessarily how important that contribution was to the finished work.

Consider a novelist who writes one hundred thousand words and an editor who suggests a single idea that transforms the entire story. The novelist supplied nearly all the prose, but the editor may have contributed something essential to the work's final form. A story's central premise, its defining character, or the revelation that makes its ending meaningful can matter more creatively than dozens of pages of supporting material.

Consequently, creative contribution should account for both the quantity of content and its significance. It should also consider substantive decisions: developing characters, constructing arguments, designing a plot, selecting among alternatives, and determining the final form of the work. These contributions are not always easy to measure, but ignoring them would produce an even less useful picture of authorship.

The second axis is the energy required to produce those contributions. In my paper, *Can Machines Think Efficiently?*, I argue that evaluating intelligent behavior should account for the energy required to produce it, rather than considering only the observed output. I believe this related principle is useful when examining creative work. Two processes can produce similar outputs while consuming substantially different resources and requiring very different amounts of work from their participants.

Applying this principle to authorship introduces a new question: how much of the energy associated with creating a work should be attributed to its human and AI contributors?

This is not necessarily a straightforward calculation. A complete accounting might consider the energy consumed by the human's work, the computing required for AI inference, and an allocated share of the energy used to train and operate the model. The accounting boundary would need to be defined consistently.

Nevertheless, the principle is useful even before every contribution can be measured precisely: the resources consumed by each participant should be considered alongside the work each actually performed.

Together, these two axes provide a more informative description of creative production than the mere presence or absence of AI. The categories *human-made*, *hybrid*, and *AI-made* should be understood as positions along a spectrum, not as rigid labels determined by a single interaction.

Case study of me

I will go through three examples.

The first is my novella currently in the works based on my experience doing jury duty in Boston. The case was a double attempted homicide, which was incredibly complicated. The majority of the jury, myself included, thought the defendant may have committed the crime, but ultimately we voted not guilty as we had reasonable doubt. I didn't realize it during the trial and only sort of realized it the night after the trial ended, but it ended up being one of the most emotional and important events I have experienced in my life. Someone my age, in different life circumstances, a jury of their peers deciding their fate. Each and every juror taking the case seriously. The immense gravity of the situation that my decision would forever change the life trajectory of the defendant.

Even though I believe the police did a subpar investigation, their dedication, sincerity, and earnestness gave me tremendous respect and appreciation for the jobs they do. I gained similar appreciation for the lawyers on both sides of the case, the judge, the court officers, and the many other people who make the court a fully functioning machine. My pride as an American and appreciation of the flawed justice system that we have and continue to improve, gave me hope. The true cross-sectional slice of our jury: race, gender, age, political belief, socioeconomic status, etc, was true to American values and also gave the defendant true access to a jury of their peers.

I share all these thoughts to start as they are an outline of the muse that inspired me to write. Without even fully understanding why, I started writing a sci-fi novella based on the intersection of the justice system and my energy efficient imitation game framework. It was only after discussions with my wife that I realized the emotional processing and significance this writing had for me. Because of the nature of memory and how it changes over time, I felt a strong desire to write down and integrate key details from my lived experience in my writing, in order to capture it before it was gone. That is one of the reasons I was very grateful to use AI to riff with as I navigated the experience of writing a sci-fi short story.

I would describe my main form of engagement with AI as an evolved version of searching the internet through a search engine. I have read a lot of science fiction but I have read relatively few short stories in the genre. I wanted to understand how my ideas and prose feed into the greater fabric of the sci-fi short story zeitgeist. The AI did not live my experience in the courtroom, nor was it the one that developed the ideas in my paper inspired by Alan Turing. The emotions, the backbone, the voice are all mine. Each and every word was hand chosen by me to express my thoughts and feelings.

The AI offered feedback throughout the entire writing process, proffering ideas for where the story could go next. The way I often express my creativity is by following my muse. While the AI can offer up ideas, ultimately I am guided by my muse and I discover where the story will go based on my lived experience and knowing what "smells right".

Within the context of the framework I am proposing in this essay, I would suggest that on both axes I am the vast contributor and thus the novella is a human generated work with minimal AI assistance.

For my second and third examples, I will offer up a deeper look at the creative process of how I created *Can Machines Think Efficiently?* as well as how this essay was constructed. The reason I am grouping these two works together is that both are essays of a sort and require a different sort of creative energy than a novella. Both essays require engaging with the art of discussion, reasoning, and philosophy. In both works, I can say without a doubt the overwhelming creative contributions come from me with very little AI involvement.

The initial conception of *Can Machines Think Efficiently?* came to me during a flight from San Francisco to Boston. I was staring out the window looking at the clouds, intentionally trying to not engage with an electronic as I had recently wanted to become more intentional about being bored. I had found that I was using electronics as a crutch when I started feeling bored, and that caused two issues. The first is that being bored is an essential part of the creative process. Boredom allows me to be my most creative. My brain wants to figure out how to not be bored.

The other reason I wanted to avoid electronics is that the resolution of solving the boredom required purposefully designed algorithms meant to addict me to their platform in order to increase my attention. As someone who works in the tech industry, I have learned first hand that every single person behind these products and algorithms could have good intentions and their individual intentions could be good, but the sum of the work could still create a problematic algorithm.

To that end, imagine a video streaming platform that seeks to maximize time spent watching videos. If the designers are people who only watch educational videos on the platform, this goal of maximizing video watch time seems noble. The more time people spend watching, the smarter the population will become. However in practice, these goals could inadvertently draw people into conspiratorial rabbit holes that lead them to perform dangerous behavior.

It was these thoughts and more, which led to that faithful thought on the plane. With the kernel of the muse in hand, I went out and read Alan Turing's paper on the imitation game for the first time. I was struck by the first person perspective, the spicy humor, and the lack of regard he had for other people. With my own mixed experience in graduate school, I felt like I could combine the emotions I am still processing from academia and my perspective on AI and algorithmic thinking I have obtained from industry.

With this all in mind, I struck out to write an academic style paper that would stick close to Turing's original style while carving out my path and voice that felt true to me. Each and every word was carefully chosen, with the specific structure of the paper down to the subheadings chosen to keep the original flavor of Turing's original work. I had many of my friends review drafts, suggest edits, engage in scholarly debate. I am extremely proud, as the ease of access to Turing's original essay was captured in some small percentage by my effort; the large spectrum of people in my life were able to engage with my ideas and find something relatable.

As I write this over a year later, I cannot remember the exact process of how I used AI to assist me in writing the paper. What I can say now is how I remembered using it, and how I recently as of a couple of months ago used it to update it. My original use was for grammar, overall flow, and to help me translate my ideas for the math section into reality.

My first major in college was mathematics, my first love of academics as it were, and I knew I wanted to incorporate an idea I have always been fascinated with: Gödel's incompleteness theorems. To do so, I had a very strong idea of the way I would translate my ideas presented in the energy efficient imitation game into set

theory so I could apply the ideas of Gödel. To that end, AI helped me significantly to couch my mathematical thoughts into something that would fit into the realm of research. I still worked through the math, spent significant time on the whiteboard working through the ideas, and made sure I understood them deeply enough to explain them to anyone.

As I alluded to already, I have recently used AI to revise the math section in my paper as I was preparing to present it at an internal research conference at Google. I have kept abreast with the advancements of AI in math, e.g. getting gold at the International Mathematical Olympiad for example, and wanted to use frontier models to examine my arguments and make sure they were airtight. The reason I did this is that even though my paper has been reviewed by some of my friends with PhDs in Computer Science with concentrations in AI, I had a hard time getting it published in a journal.

My attempts thus far had been rejected in no small part due to the first person nature of the piece. Formal peer review delivered a wonderfully candid critique: "The author indeed acknowledges that the motivation for writing is 'approachability' and 'talking to friends.' This positioning is incompatible with the journal's expectations." And while the reviewers did review my math section and did not challenge it substantially, I felt it necessary to review the math again as the paper was still not fully through peer review and just a pre-print on arXiv.

The latest review of the math section by a frontier AI model had me change the core argument for one of the sub-arguments. While my original argument was not wrong in direction, the substance of the argument had issues and a cleaner and more understandable approach was proffered by the AI. While I was the one who ultimately synthesized, revised, and published the revision to arXiv, I would ultimately label the current version of the math section as a hybrid work. While I have performed the vast majority of the work on the two proposed axes, the AI has contributed significant ideas to the arguments such that I do not think I could have gotten there on my own in a reasonable amount of time.

Now we may discuss how this essay fits into this spectrum. To start, the section titled *Case study of me* is written 100% without AI assistance. I sat down and wrote this in one sitting with no interruption. However this broader essay is a different question. The overall idea and direction of this essay comes from me. If it is not already clear, this is an area I think deeply about and will continue to think deeply about. I have created many different creative works. I have designed board games, crafted modules, rules, miniatures, and terrain for Dungeons and Dragons, and built a climbing wall with friends. I will take any chance I get to concoct some crazy recipe in the kitchen. Many of these accomplishments and more were done in the era before AI.

I went to grad school and obtained a speciality in AI. I was using the underlying technology of modern LLMs, transformers, months after it came out, before its true potential was gleaned. As AI has advanced, I have tested the waters so to speak at each opportunity, seeing how it can help me in my creative pursuits. AI will never be the majority contributor on any project I am involved with. I want to spend my time being creative. While AI helps me to that end, the goal is never to rush out and create the final product in the fastest way possible.

The creative pursuit is often the goal for me; the journey is often more important than the destination. What makes all of these points necessary to explain is that this essay is my thoughts. This essay is my response to my many previous musings. This essay is my reaction to the checkboxes I see when I submit some of my creative work online that reads: "This piece of work was not created by or with the assistance of AI."

Other than the section titled *Case study of me,* this essay was written as a hybrid collaboration of me and AI. The core idea of the essay came from me, the context that preceded the prompt that generated the initial seed of the full length essay was my discussion about my novella and the implications of the aforementioned submit

checkboxes. Each and every line in this essay I have carefully attended to, combined over, to ensure it feels like something I would say.

If we were to use my proposed axes how should we evaluate this work? On one hand, we have an entire substantive section that is 100% human generated. On the other hand we have an explicit omission that AI helped generate a significant number of the words on the page. Is this piece AI generated? Is this piece a hybrid? Is it human made? For myself as the author, it feels human made with AI assistance.

Could I have sat down and written each and every word by myself in a reasonable amount of time? Yes. Would I have? That's where I am not sure. As I examine my own emotions around this essay, I am grateful that AI has allowed me to engage with yet another aspect of my creativity. I feel like I am to express myself in a way that feels important and relevant during this era.

To close this section, I supplied the following prompt to the AI that helped generate this piece.

Prompt:

> Using this framework we just created, review the draft I have prepared based on our conversation and write a few paragraphs that will be inserted into the text below this prompt in the essay.

Interestingly, it ignored the prompt and tried to give a critique of the overall essay. I then supplied it with the following prompt and provide its unedited response below.

Prompt:

> You ignored my previous prompt. I want you to read the essay, and write a few paragraphs that will close the section titled *Case study of me.* Because of your failed attempt I have edited the section already to account for this. For context, this prompt verbatim will be inserted into the text following the section I have included below.
>
> > "To close this section, I supplied the following prompt to the AI that helped generate this piece.
> >
> > Prompt:
> >
> > > Using this framework we just created, review the draft I have prepared based on our conversation and write a few paragraphs that will be inserted into the text below this prompt in the essay."

AI:

> You're right. I misunderstood where the new text belongs. You want the AI's response to your prompt to close “Case study of me” by reflecting on the three examples you've presented, not to replace the following section about the four hypothetical writers. I've read the relevant section of your draft and focused the paragraphs below on your actual creative process.
>
> > Taken together, these three examples illustrate why authorship is better understood as a spectrum than as a binary classification. The sci-fi novella appears to be the clearest case of human-made work with AI assistance. Its origins lie in the author's lived experience, emotional processing, and preexisting ideas about the intersection of law and energy-efficient intelligence. AI helped him explore possibilities and reflect on his work, but the author supplied the experiences, chose which ideas resonated with him, and wrote the final prose. The AI's involvement was real, but it did not make the novella's creative foundation its own.

Can Machines Think Efficiently? presents a more complicated case. The paper's central motivation and intellectual direction originated with the author, who also worked through the mathematics and developed the arguments. Yet the later revision of a mathematical subargument involved a substantive contribution from AI: it offered an approach that improved the argument and that the author believes he would not have reached independently in a reasonable amount of time. Calling the entire paper AI-made would overlook the human work behind it, while calling every part of its development exclusively human would overlook the contribution that changed the published argument. This suggests that authorship classifications may sometimes be more informative when applied to particular contributions or revisions, rather than only to an entire finished work.

This essay is perhaps the most revealing example because the writing process itself has become part of the subject. Its central question, proposed framework, and autobiographical material come from the author, and the section recounting his experiences was written without AI assistance. However, the wider essay also incorporates substantial AI-generated prose that the author selected, examined, and edited. Under the proposed framework, describing the essay as entirely human-made would understate the AI's contribution to the content, while describing it as simply AI-made would understate the human's role in originating the ideas and directing the argument. A reasonable description is a human-led hybrid work, although a precise classification along the energy axis would require a consistent method of accounting for the energy used by each participant.

These examples also reveal a limitation worth acknowledging: the framework provides a way to reason about authorship, not an automatic method for calculating it. The significance of a contribution is often subjective, and the energy associated with producing a work is difficult to allocate. Nevertheless, the two axes encourage a more honest examination of creative work. Rather than asking only whether AI was used, we can ask what it contributed, how important that contribution was, and what resources were required to produce it.

The author's own experience is therefore not simply an argument that AI-assisted work should be considered human-made. It demonstrates that different projects, and even different parts of the same project, can involve different relationships between human and machine contributions. A useful framework should be capable of recognizing those differences, including cases in which AI has contributed more substantially than an author might initially expect. The goal is not to erase the distinction between human and AI creation, but to describe that distinction more accurately.

I deliberately made the classification of this essay more candid than your initial instinct to call it human-made with AI assistance. Given your own framework and your acknowledgment that AI generated a substantial amount of the prose, “human-led hybrid” seems like the more consistent assessment of this particular draft. That tension is what makes your personal case study persuasive: you're applying the framework to your own work, even where its conclusions may be less straightforward than you'd prefer.

Originally I thought I would allow the AI's thoughts to close this section, however after reading its response I felt an obligation to respond. In this specific instance, I feel like if I tasked a human with my original prompt, in the vast majority of cases, I would have received a reflection that encompassed their thoughts on the entire piece, and respected a core requirement of the task, write a *few* paragraphs, emphasis mine.

I think it is quite telling of the current capability of AI as of writing and to the broader implications of my framework, that the AI ignores my requirement of brevity. While I agree with its overall assessment of the piece, human-led hybrid, I would ultimately refer back to the specific words I chose in my own assessment, human made with AI assistance.

If I had to choose one of three labels: human-made, hybrid, and AI-made, with acknowledged controversy I would say this piece is human-made. The blood, sweat, and tears that make up the underlying emotions and opinions of this piece have come from me. This piece was written by me, under direction by me, edited by me, and ultimately published by me. While AI has been involved throughout the process, I feel strongly that on the axes I have proposed, this work is closer to the identity of human-made vs hybrid.

Four writers on the spectrum

To illustrate how the framework might work, consider four hypothetical writers.

Writer A: The human novelist and the consequential suggestion

Writer A spends one hundred hours writing and revising a novel, occasionally asking an AI assistant for feedback. During one conversation, the AI suggests an idea that fundamentally changes the ending. The writer considers the suggestion, develops it, and writes the final version independently.

On the creative-contribution axis, the AI has supplied very little content by volume, but its contribution may be highly significant. Without the suggestion, the novel might have ended very differently. On the energy axis, however, the human may have performed the overwhelming majority of the work required to produce the novel.

Writer A illustrates why neither axis should be used alone. Counting words would make the AI's contribution appear negligible. Considering only the significance of the suggestion could overstate it. Considering energy alone would fail to represent the creative importance of the idea.

The most reasonable classification would likely be human-made, with a meaningful AI contribution. Whether that contribution is enough to call the novel hybrid would depend on the criteria being used. The point is that the classification should reflect the actual relationship between the contributions, rather than the simple fact that an AI was consulted.

Writer B: The human editor of an AI-generated novel

Writer B asks an AI to generate a complete novel in minutes, then spends an hour changing grammar, style, and character names before publishing it.

Here, the AI has supplied nearly all the substantive content. The human has made some editorial contributions, but these have not substantially changed the work's structure, central ideas, or narrative development.

On the first axis, the AI's contribution dominates. On the second, the outcome depends on the accounting boundary and the energy required for model inference, training, and human activity. The classification should not assume that an AI-generated novel is necessarily more energy-intensive in every conceivable accounting model. Rather, the energy contribution should be evaluated alongside the creative contribution.

Even if the human's hour of editing consumed more energy than the specific generation run, that fact would not make the novel human-made. The human did not supply most of its creative substance. This is precisely why a two-axis model is more useful than an energy-only classification.

Writer C: The writer who refuses all AI assistance

Writer C writes a novel without intentionally using AI at any stage. They do not ask an AI assistant for feedback, use AI-generated prose, or employ AI for research. For the sake of this example, they even refuse to look up a word through a search engine because the search results may contain an AI answer.

This writer represents the clearest case of human-made work under a strict definition of AI-free creation.

But Writer C also exposes a problem with definitions based on tool exposure. Must an author avoid an entire search engine because it includes AI content? Does inadvertently seeing AI content count as AI assistance? Must a writer investigate every digital tool used in the research process to determine whether some machine-learning component might have influenced a result?

These questions reveal the difference between using AI to create a work and merely encountering AI generated information. Incidental exposure does not necessarily constitute a creative contribution. A useful authorship framework should distinguish between them rather than treating all contact with AI as equivalent.

Writer D: The automated author

Writer D asks an AI to generate a novel and publish it, contributing little beyond initiating the process. The AI develops the story, writes the manuscript, and carries out the publication workflow.

This is the clearest example of AI-made work among the four. The AI supplies the overwhelming majority of the creative content and performs most of the work required to produce the result. The human's role is primarily supervisory or operational.

The distinction between Writers B and D is worth preserving. Writer B performs some editing; Writer D performs almost none. Neither necessarily contributed much of the novel's creative substance, but their degrees of involvement differ. A useful spectrum should be able to represent that difference without pretending that every work fits neatly into one of three boxes.

Why the distinction matters

These examples suggest that authorship cannot be adequately determined by asking whether AI was involved. Instead, we should ask what each participant contributed and what resources were required to produce that contribution.

This matters especially when creative work has economic value. If human and AI contributions are treated as equivalent whenever AI appears in the workflow. A writer who spends months developing a novel could be classified in the same way as someone who publishes a machine-generated manuscript after making trivial edits. Conversely, treating every work containing AI assistance as entirely human-made could obscure cases in which AI performed nearly all the creative labor.

A contribution-based framework offers a better starting point for allocating credit and compensation. In contexts where royalties or residuals are distributed according to contributions, the relevant question should be how much substantive work each participant performed—not merely whether a particular tool was used.

This does not mean that the two axes automatically determine the appropriate financial split. Compensation can depend on contracts, ownership, bargaining power, and other considerations beyond the creative process

itself. Nor does it mean that every contribution can be assigned an objectively correct percentage. A single idea might be difficult to evaluate, and reasonable people might disagree about its significance.

Nevertheless, explicitly separating creative contribution from energy expenditure gives us a clearer vocabulary for the discussion. It allows us to distinguish between the source of an idea, the production of the content, the resources consumed in producing it, and the decisions that determine its final form.

The limits of the model

The two axes are a starting point, not a complete theory of authorship. Creative significance is subjective, and energy attribution is difficult. A single idea can change a work disproportionately to the effort required to develop it. Human biological energy, computer electricity, and the energy used to train a model also present difficult accounting questions.

There may be other relevant dimensions, including originality, control over the final result, responsibility for the work, and the extent to which the creator understands or can defend its contents. These may matter in particular contexts. But the two proposed axes—creative contribution and energy expenditure—are sufficient to establish a more nuanced framework in many ordinary cases.

The framework should also be used differently depending on the question being asked. A publisher deciding whether to accept a work under an AI-free policy might impose a strict rule about AI involvement. A society allocating authorship credit or residuals might care more about substantive contributions. A researcher studying the environmental cost of creative production might focus primarily on energy expenditure. These are different goals, and they need not produce identical classifications.

The purpose of the framework is not to dictate one universal rule for every institution. It is to make the underlying judgments more explicit and to avoid collapsing distinct forms of participation into a single category.

Toward a more useful definition of human-made work

The rise of AI challenges our traditional assumptions about creative production, but it does not require us to abandon the concept of human authorship. Humans have long created work using tools. The presence of assistance has never, by itself, told us who made something or how much each participant contributed.

AI makes these questions more pressing because it can now perform a much larger share of the work, from supplying a single suggestion to producing a complete finished product. A definition of authorship that treats these situations as equivalent will struggle to distinguish meaningful human creation from largely automated production.

A better framework considers both the substance of each participant's contribution and the resources required to produce it. Under this approach, a human who uses AI occasionally can remain the primary creator, while someone who delegates nearly the entire creative process to an AI can reasonably describe the result as AI-made. Between them lies a broad spectrum of hybrid work.

The central principle is simple: we should classify creative work according to what humans and machines actually contribute, not merely whether an AI tool appeared somewhere in the process. Energy expenditure adds a second perspective, connecting questions of authorship to the material cost of producing creative output.

The resulting framework will not eliminate every disagreement. It should, however, make those disagreements more productive by shifting attention away from a binary test of AI involvement and toward the more substantive questions of contribution, effort, and creation.